



Sequential processing of measurements

- The standard Kalman filter is an efficient recursive algorithm.
- However, under some conditions, we can improve its speed by rewriting some of its equations.
- One of the computationally intensive operations in the Kalman filter is the matrix inverse operation in Step 2a: $L_k = \Sigma_{\tilde{x},k}^{-1} C_k^T \Sigma_{\tilde{z},k}^{-1}$.
- Inverting $\Sigma_{\tilde{z},k}^{-1}$ via Gaussian elimination (the most straightforward approach), it is an $\mathcal{O}(m^3)$ operation, where m is the dimension of the measurement vector.
- If there is a single sensor, this matrix inverse becomes a scalar division, which is an $\mathcal{O}(1)$ operation—very fast; otherwise it is slow.
- Therefore, if we can break the m measurements into m single-sensor measurements and update the Kalman filter that way, there is opportunity for significant computational savings.



Start by assuming v_k are uncorrelated

- We start by assuming that the sensor measurements are uncorrelated. This implies that:

$$\Sigma_{\tilde{v}} = \text{diag} \left[\sigma_{\tilde{v}_1}^2, \dots, \sigma_{\tilde{v}_m}^2 \right].$$

- We will use colon “:” notation to refer to the measurement number. For example, $z_{k:1}$ is the measurement from sensor 1 at time k .
- Then, the measurement is

$$z_k = \begin{bmatrix} z_{k:1} \\ \vdots \\ z_{k:m} \end{bmatrix} = C_k x_k + v_k = \begin{bmatrix} C_{k:1}^T x_k + v_{k:1} \\ \vdots \\ C_{k:m}^T x_k + v_{k:m} \end{bmatrix},$$

where $C_{k:1}^T$ is the first row of C_k (for example), and $v_{k:1}$ is the sensor noise of the first sensor at time k , for example.



The i th gain matrix

- We will consider z_k to be sequence of scalar measurements $z_{k:1} \dots z_{k:m}$ and update the state estimate and covariance estimates in m steps.
- We initialize the measurement update process with $\hat{x}_{k:0}^+ = \hat{x}_k^-$ and $\Sigma_{\tilde{x},k:0}^+ = \Sigma_{\tilde{x},k}^-$.
- Consider the measurement update for the i th measurement, $z_{k:i}$:

$$\begin{aligned} \hat{x}_{k:i}^+ &= \mathbb{E}[x_k | \mathbb{Z}_{k-1}, z_{k:1} \dots z_{k:i}] \\ &= \mathbb{E}[x_k | \mathbb{Z}_{k-1}, z_{k:1} \dots z_{k:i-1}] + L_{k:i} (z_{k:i} - \mathbb{E}[z_{k:i} | \mathbb{Z}_{k-1}, z_{k:1} \dots z_{k:i-1}]) \\ &= \hat{x}_{k:i-1}^+ + L_{k:i} (z_{k:i} - C_{k:i}^T \hat{x}_{k:i-1}^+). \end{aligned}$$

- Generalizing from before:

$$L_{k:i} = \mathbb{E}[\tilde{x}_{k:i-1}^+ \tilde{z}_{k:i}^T] \Sigma_{\tilde{z},k:i}^{-1}.$$



State and error-covariance update

- Next, we recognize that the variance of the innovation corresponding to measurement $z_{k:i}$ is:

$$\Sigma_{\tilde{z}_{k:i}} = \sigma_{\tilde{z}_{k:i}}^2 = C_{k:i}^T \Sigma_{\tilde{x},k:i-1}^+ C_{k:i} + \sigma_{\tilde{v}_i}^2.$$

- The corresponding gain is $L_{k:i} = \Sigma_{\tilde{x},k:i-1}^+ C_{k:i} / \sigma_{\tilde{z}_{k:i}}^2$, and the updated state is:

$$\hat{x}_{k:i}^+ = \hat{x}_{k:i-1}^+ + L_{k:i} [z_{k:i} - C_{k:i}^T \hat{x}_{k:i-1}^+]$$

having covariance: $\Sigma_{\tilde{x},k:i}^+ = \Sigma_{\tilde{x},k:i-1}^+ - L_{k:i} C_{k:i}^T \Sigma_{\tilde{x},k:i-1}^+.$

- The covariance update can be implemented as:

$$\Sigma_{\tilde{x},k:i}^+ = \Sigma_{\tilde{x},k:i-1}^+ - \Sigma_{\tilde{x},k:i-1}^+ C_{k:i} C_{k:i}^T \Sigma_{\tilde{x},k:i-1}^+ / (C_{k:i}^T \Sigma_{\tilde{x},k:i-1}^+ C_{k:i} + \sigma_{\tilde{v}_i}^2).$$

- The final measurement update gives $\hat{x}_k^+ = \hat{x}_{k:m}^+$ and $\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k:m}^+.$



Sequentially processing correlated measurements

- This process must be modified to accommodate situations where sensor noise is correlated among the measurements.
- Assume that we can factor the matrix $\Sigma_{\tilde{v}} = S_v S_v^T$, where S_v is a lower-triangular matrix (for symmetric positive-definite $\Sigma_{\tilde{v}}$, we can).
 - Recall that the factor S_v is a kind of a matrix square root, known as the “Cholesky” factor of the original matrix.
 - In Octave, `Sv = chol(SigmaV, 'lower');`
 - Be careful: Octave’s default answer (without specifying “lower”) is an upper-triangular matrix, which is not what we’re after.
- The Cholesky factor has strictly positive elements on its diagonal (positive eigenvalues), so is guaranteed to be invertible.



Decorrelating correlated measurement noise

- Consider a modification to the output equation of a system having correlated measurements:

$$z_k = C x_k + v_k$$

$$\bar{z}_k = S_v^{-1} z_k = S_v^{-1} C x_k + S_v^{-1} v_k = \bar{C} x_k + \bar{v}_k.$$

- Note that I am not using the “bar” symbol to indicate the mean of a quantity here.
- Instead, $\bar{z}_k$ is a computed output, similar in interpretation to measured output z_k .
- Consider now the covariance of the modified noise input $\bar{v}_k = S_v^{-1} v_k$:

$$\begin{aligned} \Sigma_{\bar{v}_k} &= \mathbb{E}[\bar{v}_k \bar{v}_k^T] \\ &= \mathbb{E}[S_v^{-1} v_k v_k^T S_v^{-T}] = S_v^{-1} \Sigma_{\tilde{v}} S_v^{-T} = I. \end{aligned}$$

- Therefore, we have identified a transformation that de-correlates (and normalizes) measurement noise.



Revised measurement-update process

- Using this revised output equation, we use the prior method.
- We start the process with $\hat{x}_{k:0}^+ = \hat{x}_k^-$ and $\Sigma_{\tilde{x},k:0}^+ = \Sigma_{\tilde{x},k}^-$.
- The Kalman gain is $\bar{L}_{k:i} = \frac{\Sigma_{\tilde{x},k:i-1}^+ \bar{C}_{k:i}^T}{\bar{C}_{k:i}^T \Sigma_{\tilde{x},k:i-1}^+ \bar{C}_{k:i}^T + 1}$ and the updated state is:

$$\begin{aligned}\hat{x}_{k:i}^+ &= \hat{x}_{k:i-1}^+ + \bar{L}_{k:i} [\bar{z}_{k:i} - \bar{C}_{k:i}^T \hat{x}_{k:i-1}^+] \\ &= \hat{x}_{k:i-1}^+ + \bar{L}_{k:i} [(\mathcal{S}_v^{-1} z_k)_i - \bar{C}_{k:i}^T \hat{x}_{k:i-1}^+],\end{aligned}$$

having covariance:

$$\Sigma_{\tilde{x},k:i}^+ = \Sigma_{\tilde{x},k:i-1}^+ - \bar{L}_{k:i} \bar{C}_{k:i}^T \Sigma_{\tilde{x},k:i-1}^+.$$

- The final measurement update gives $\hat{x}_k^+ = \hat{x}_{k:m}^+$ and $\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k:m}^+$.



LDL updates for correlated measurements

- An alternative to the Cholesky decomposition for factoring the covariance matrix is the LDL decomposition:

$$\Sigma_{\tilde{v}} = \mathcal{L}_v \mathcal{D}_v \mathcal{L}_v^T,$$

where $\mathcal{L}_v$ is lower-triangular and $\mathcal{D}_v$ is diagonal (with positive entries).

- In MATLAB (not available in Octave), `[L,D] = ld1(SigmaV);`
- The Cholesky decomposition is related to the LDL decomposition via: $S_v = \mathcal{L}_v \mathcal{D}_v^{1/2}$.
- We can use the LDL decomposition to perform a sequential measurement update.¹
 - A computational advantage of LDL over Cholesky is that no square-root operations are required. (We can avoid finding $\mathcal{D}_v^{1/2}$.)
 - A pedagogical advantage of Cholesky is that we use it elsewhere.

¹For example, see: Dan Simon, *Optimal State Estimation: Kalman, H_∞ , and Nonlinear Approaches*, Wiley Interscience, Hoboken, New Jersey, 2006.



Summary

- When the system we are monitoring has multiple outputs, it is possible to speed up the Kalman filter by processing measurement sequentially.
- If the sensor noises are uncorrelated, the approach directly updates the state estimate and its covariance using one sensor at a time.
- If the sensor noises are correlated, the measurements must be jointly preprocessed by multiplying by a decorrelating Cholesky factor; then, the processed measurements can be used to update the state estimate and its covariance one at a time.